

Mathematics A

Unit 2 • Chapter OL-T23 Classified

Cross-session • chapter-organised candidate

9 classified items • 27 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

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Contents

Unit 2 • Unit 2 • Chapter OL-T23 • 9 items • 27 marks

Chapter	Items	Marks	Starts
01. OL-T23 Algebraic Proof	9	27	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Algebraic Proof

Unit 2 • 27 marks • 9 item(s)

June 2024 | 4MA1-2H Q14 | 3 marks

Prove two algebraic forms are identically equivalent

January 2022 | 4MA1-2H Q18 | 3 marks

Fixed remainder modulo a divisor

January 2020 | 4MA1-2H Q17 | 3 marks

Consecutive integer parity identity

June 2021 | 4MA1-1H Q17 | 3 marks

Consecutive even/odd sequence proof

November 2021 | 4MA1-1H Q19 | 3 marks

Derive a general geometric algebra formula

January 2023 | 2H Q17 | 4 marks

Product or sum of arbitrary odd/even numbers

November 2025 | 1H Q5 | 3 marks

Expression built from a known multiple

November 2025 | 1H Q17 | 3 marks

Product or sum of arbitrary odd/even numbers

November 2025 | 2H Q16 | 2 marks

Derive a general geometric algebra formula

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Question 19

4MA1-2H Q14

MODULAR UNIT 2

3 marks

- 14 (a) Show that $3y(2y + 5)(y + 7)$ can be written in the form $ay^3 + by^2 + cy$ where a, b and c are integers.

(3)

WORKING SPACE

Question 30

4MA1-2H Q18

MODULAR UNIT 2

3 marks

- 18 Prove that when the sum of the squares of any two consecutive odd numbers is divided by 8, the remainder is always 2
Show clear algebraic working.

WORKING SPACE

- 17 Prove that the difference between two consecutive square numbers is always an odd number.
Show clear algebraic working.

YOUR WORKING

- 17 Using algebra, prove that, given any 3 consecutive even numbers, the difference between the square of the largest number and the square of the smallest number is always 8 times the middle number.

YOUR WORKING

19 $ABCED$ is a five-sided shape.

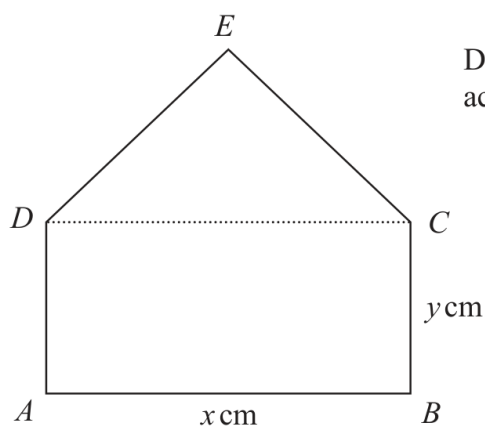


Diagram **NOT**
accurately drawn

$ABCD$ is a rectangle.

CED is an equilateral triangle.

$$AB = x \text{ cm} \quad BC = y \text{ cm}$$

The perimeter of $ABCED$ is 100 cm.

The area of $ABCED$ is $R \text{ cm}^2$

(a) Show that $R = \frac{x}{4} (200 - [6 - \sqrt{3}]x)$

(3)

(b) (i) Find the value of x for which R has its maximum value.

Give your answer in the form $\frac{p}{q - \sqrt{3}}$ where p and q are integers.

$$x = \dots\dots\dots$$

(2)

(ii) Explain why the maximum value of R is given by this value of x .

.....

.....

.....

(1)

YOUR WORKING

.....

.....

.....

17 Given that $8\sqrt{m} + \sqrt{49m} - \sqrt{9m} = k\sqrt{m}$
where k is an integer and m is a prime number,

(a) work out the value of k

$$k = \dots\dots\dots (1)$$

(b) Show that $\frac{5 - \sqrt{18}}{1 - \sqrt{2}}$ can be written in the form $a + b\sqrt{2}$

where a and b are integers.
Show each stage of your working clearly.

(3)

(Total for Question 17 is 4 marks)

YOUR WORKING

5 Show that $3\frac{5}{7} + 1\frac{2}{3} = 5\frac{8}{21}$

(Total for Question 5 is 3 marks)

YOUR WORKING

17 Show that $\frac{2\sqrt{7} + 2}{\sqrt{7} - 3}$ can be written in the form $a - \sqrt{b}$ where a and b are integers.

Show your working clearly.

(Total for Question 17 is 3 marks)

YOUR WORKING

16 Use algebra to show that $0.3\dot{7}\dot{8} = \frac{25}{66}$

(Total for Question 16 is 2 marks)

YOUR WORKING
